Sarasavi Library Software Project

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1. Group Member & Work Division Details

Photo	Student Name	Index Number	NIC	**Contribution / Part Done**
	G.H Ayon Dilshan Amarashige	C100043	200622304712	Developed the View Users form to display the member list and Developed the Library Home form. Developed the Book Inquiry form and its search functionality.
	Duwindi Wijesinghe	C102406	200155602426	Developed the User Operations form, including the UI and the back-end C# logic for Save, Edit, and Delete functions. Create Project report.
	W.M.Nithya Sathsarani Bandara	C38737	200450610908	Designed and implemented the complete SQL Server database schema.
	R.M.Madhavi Nimeshika Rathnayaka	C104978	200185600904	Developed the back-end logic for: Loan Books. Developed the back-end logic for forms: Return Books, and Reserve Books.

2. Abstraction

A complete library management system called the “Sarasavi Library Software Project” is a desktop-based library software program that is meant to modernize and simplify the traditional library operations. The system is developed with the help of C# windows forms as user interface and Microsoft SQL server as the backend database and is built on the best technologies to make sure it is stable, responsive and manages the data efficiently on the windows platform.

The visually appealing theme of the application in the form of a chalkboard is one of its unique peculiarities. The design option is a combination of a traditional academic environment and the current digital features that add a modern appeal to the workspace, providing the librarian with a user-friendly and interacting work experience with the professional usability.

The system is organized based on a highly structured database schema of five tables as Users, Books, Copies, Loans, and Reservations. This architecture does not only maintain master records of book titles and library members but also maintains a record of individual physical copies of a book. The use of separate copy management allows the application to control inventory and monitor the availability of certain items correctly, which will allow managing the resources.

In terms of functionality, the program comprises a holistic selection of administrative solutions that facilitate the entire working process of a library. They consist of secure authentication system to protect data, full CRUD (Create, Read, Update, Delete) of the books and users, and advanced transactions that facilitate the circulation operations like issuing loan, handling returns and reservations.

The Sarasavi Library Software Project, in general, eliminates paper-based and error prone and manual procedures with an efficient, structured, and reliable digital solution, greatly enhancing the accuracy, productivity, and general efficiency in the management of the library.

3. Objective of the Project

The main goals of the project will be presented as follows:

- To come up with a safe, centralized, and effective digital system that will be able to oversee all the central operations of the library under one integrated system.
- Automating the full lifecycle of library inventory, including adding new titles of books, receiving single copies, tracking circulation operations and the exclusion of old and damaged materials.
- To implement a systematic and well-organized way of managing and updating the records of library members.
- To provide a backend logic to enforce library policies, including borrowing limits, user-type access privileges, as well as other circulation policies.
- To offer a high-tech and convenient search experience that helps to find the information about the book quickly and stay informed about its availability.
- To be able to manage the reservation process and notify the librarian about the pending requests in the case when the books are returned.

All these objectives have a common goal of increasing efficiency in operations, compliance in policies and general management of library resources.

4. Tools and Technologies Used

Category	**Tools and Technology**
Frontend	Windows Forms (C#)
Backend	C#.NET
Database	SQLServer
IDE	Visual Studio

5. Functional Requirements

- **Authentication and Security**

  The system shall be protected by a secure login interface. Access to the administrative functions shall be granted only upon successful authentication using the following credentials:

  - Username: **admin**
  - Password: **12345**

- **Administrative Functionalities**

  The system shall provide the administrator with complete Create, Read, Update, and Delete (CRUD) capabilities for managing both book records and user records.

- **Automatic Copy Generation**

  When a new book title is entered into the system, it shall automatically create the specified number of individual copy records and store them in the database.

- **User Management Interface**

  The application shall include a dedicated interface that enables the administrator to view a comprehensive list of all registered users.

- **Book Inquiry Module**

  The system shall provide a search functionality that allows users to locate books using:

  - Book ID
  - Partial title
  - Partial author name

- **Business Rules Enforcement**

  The system shall enforce the following operational policies:

  - Visitors shall not be permitted to borrow or reserve books.
  - Books categorized as reference-only shall not be available for loan.
  - Members shall be limited to a maximum of five active book loans at any given time.

6. System Design

- ### Use Case Diagram

The proposed library management application is designed around a single primary actor, namely the **Librarian**. The Librarian is responsible for managing and controlling all system functionalities. Access to the system is granted only after successful authentication. Once logged in, the Librarian can perform the following major use cases:

1. **User Operations**

   This use case enables the Librarian to manage user accounts within the system. It includes:

   - Registering new users
   - Editing existing user profiles
   - Deleting user records
2. **Book Operations**

   This use case allows the Librarian to manage book records and their copies. It includes:

   - Registering new book titles
   - Editing book information
   - Deleting book titles and related copy records
3. **Book Inquiry**

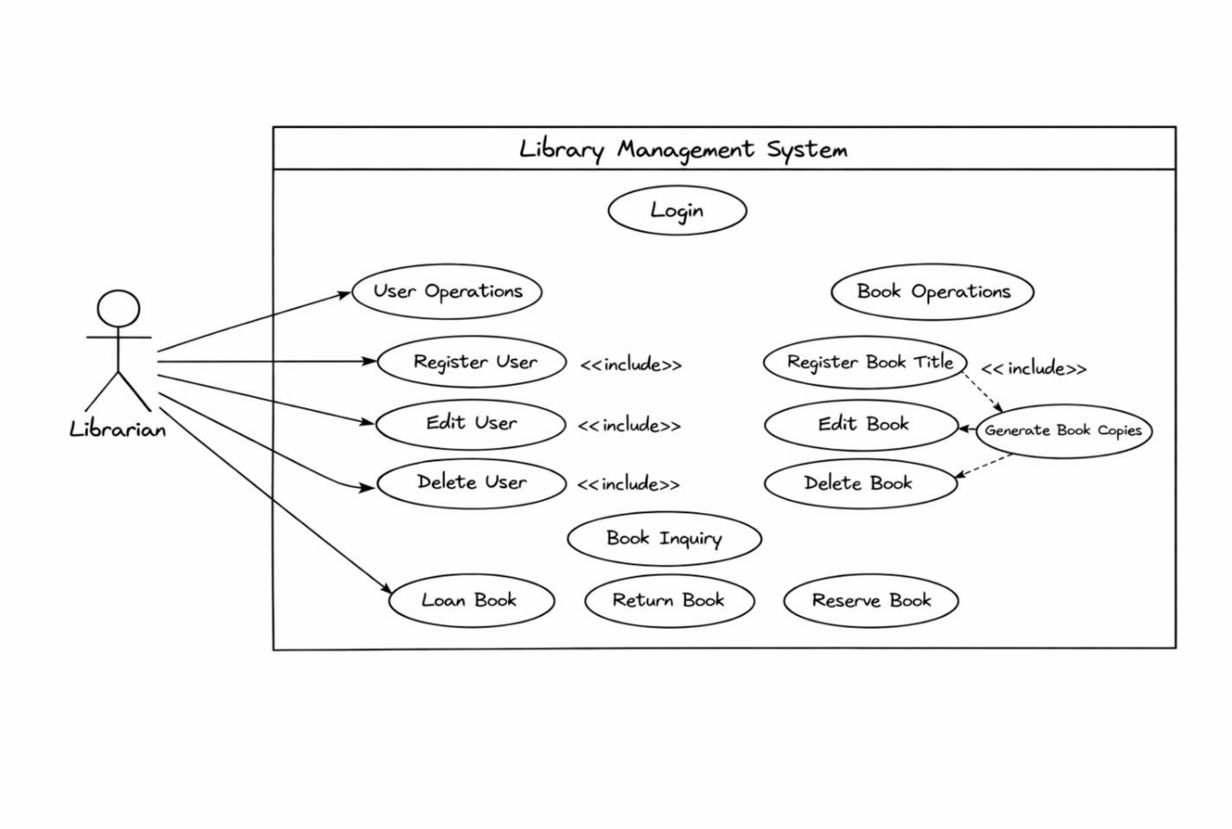
   This use case provides the Librarian with the ability to search and view details of books available in the library collection.
4. **Loan, Return, and Reservation Management**

   This use case enables the Librarian to handle all circulation-related transactions, including:

   - Issuing books to members

- Processing book returns
- Managing book reservations

In the Use Case Diagram, the **Librarian** actor is connected to all the above use cases, indicating full system access after authentication.



- ER Diagram

The database design comprises of five interdependent tables namely; Users, Books, Copies, Loans and Reservations. The schema is also normalized fully to assure data integrity, reduce redundancy and facilitate operation of libraries effectively.

Users Table

This table keeps all the information regarding the library patrons, their personal and membership information. Every record has a unique identification of a registered user into the system

Books Table

The Books table contains the base (master) data of a particular book title, including book title name, book author, publisher and the category of a book. The title of every book is registered once in this table.

Copies Table

Copies table is the physical copies of single book titles.

Identification of each copy is made.

A foreign key (book id) creates a one to many relationships between the books and copies tables, with a single physical book title having several physical copies.

Loans Table

Loans table keeps the record of borrowing transactions.

It connects a particular User with a particular Copy.

This table monitors the date of loans, date of returns and loan status.

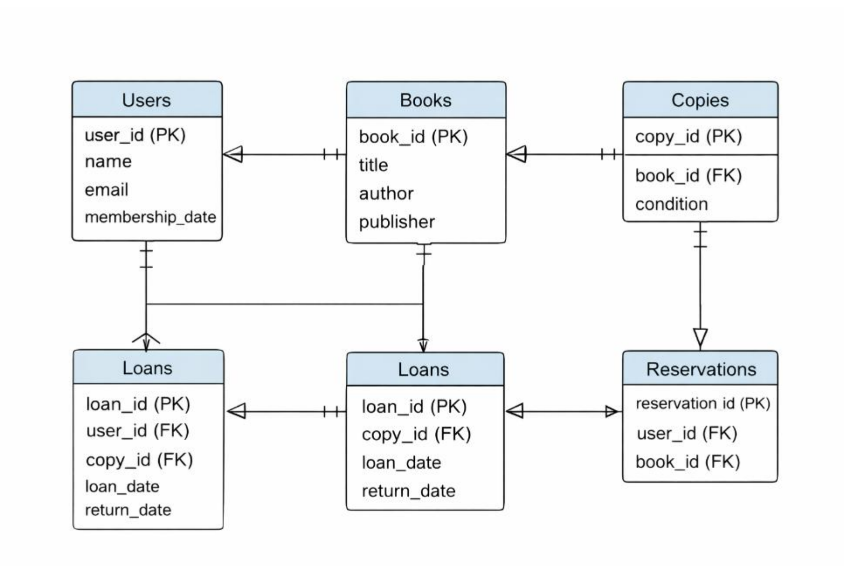
Reservations Table

Reservation table deals with book reservation requests.

It connects a User to a book title of Books table.

As opposed to Loans, reservations are linked to the master book record and not a particular copy.

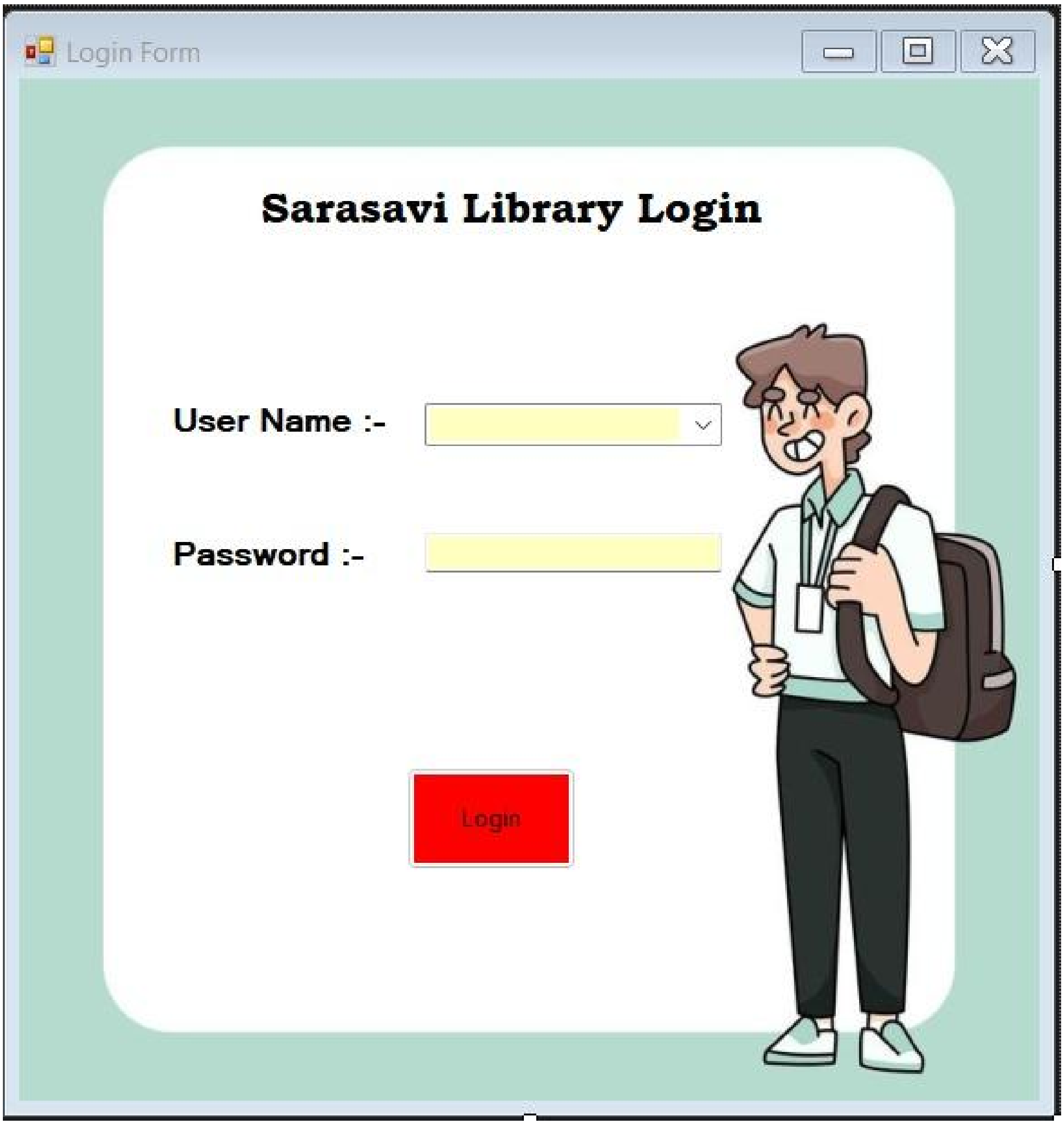
This relational design composed of order is to provide the appropriate enforcement of business rules and help manage the circulation activities effectively



7. Project UI and Explanation

The user interfaces (UI) are explained one by one below. The elements (buttons, etc.) used in each user interface (UI) and the functions performed by them are explained separately.

- Login Form



The secure entry point to the Library Management System is the Login Form. It ensures that access to the system is limited to approved staff to the features and administrative capabilities of the system.

User Interface Components and Functionalities.

Username and Password Fields.

These input fields enable the administrator to input his or her login credentials. The field of password is masked to ensure confidentiality and improve the security.

Login Button

Upon clicking the button of the Log-in, the system compares the stated credentials with the defined authentication data.

In the case of the validity of the credentials the system will allow access and redirections to the main Library Home interface.

In case the credentials are invalid an access is denied and a suitable message of error is shown.

This type is essential in ensuring the security of the systems as well as avoiding unauthorized entry.

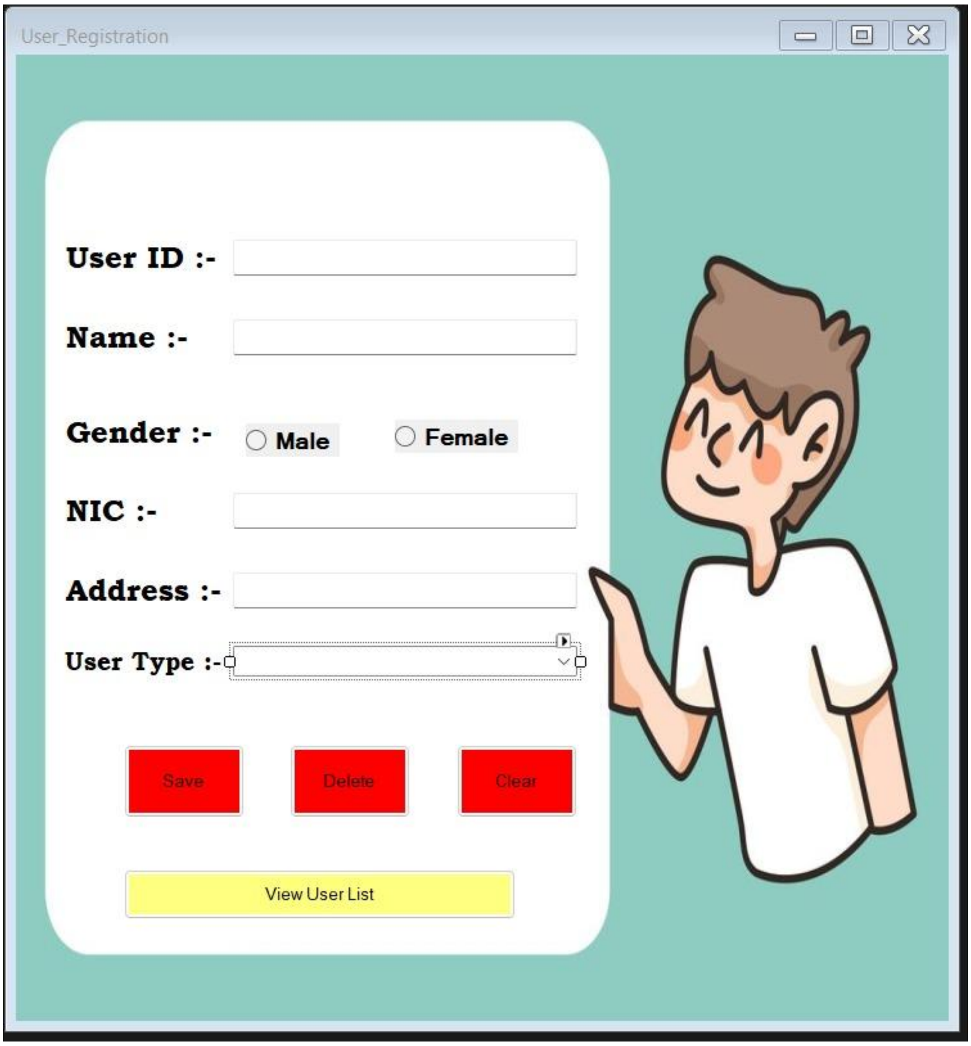
- Library Home Form

The Library Management System has a central control panel which is called the Main Dashboard. It gives direct access to all the six main features of the app in a well-organized and user-friendly interface.

Components and Functionalities of User Interface.

Navigation Buttons (User Operations, Book Operations, etc.) The dashboard has six large action buttons, each of which signifies a key system feature. Every button serves as a control button. The system launches the form in accordance to that particular operation when it is selected. The administrator can easily access and control various modules of the system by these buttons. The dashboard is structured to simplify workflow with all key features being centrally accessed by the dashboard.

- User Operations Form



User Management Form offers an overall interface in managing patron records in libraries. It allows the administrator to do complete user information management in the system.

Functionalities and User Interface Components.

User Information Fields

The form contains the input templates like User ID, Name and NIC among other details. These areas enable the administrator to add and update the patron information correctly.

Save Button

The Save button will create a new user record into the Users table through an SQL INSERT command.

Edit Button

Edit option changes the information of an existing user. It does an SQL UPDATE that is defined by the given User ID.

Delete Button

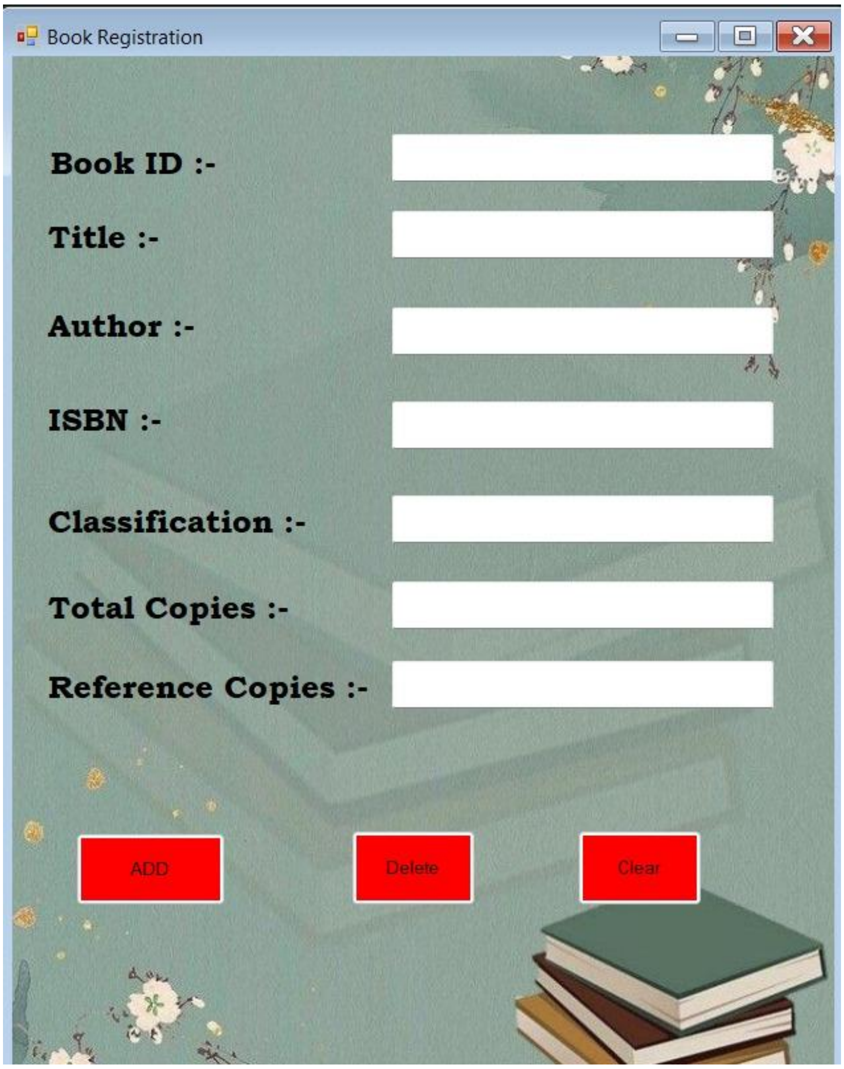
The Delete option eliminates the user record that is selected out of the database. It performs SQL DELETE on the user record to remove both the user record and related transaction records (a loan or reservation).

View User List Button

With this functionality, a new interface opens that shows a full list of all registered library patrons so that one can easily look and refer.

It is a centralized form where all activities pertaining to the users are centralized and hence efficient and accurate patron data management takes place.

- Book Operations Form



Book Management Form is used as a point of control to manage and keep track of the books available in the library. It facilitates an administrator to control the titles of books and copies of the same.

User Interface Construction and Things.

Book Information Fields

The form incorporates the input fields like Book ID, Title, Author, Publisher and other pertinent details. These areas enable the administrator to add and modify all the information of the book title.

Save Button

A new record that is inserted into the Books table is the save function. When successful, the system would automatically create the number of corresponding records in the Copies table to reflect individual physical copies of the book.

Edit Button

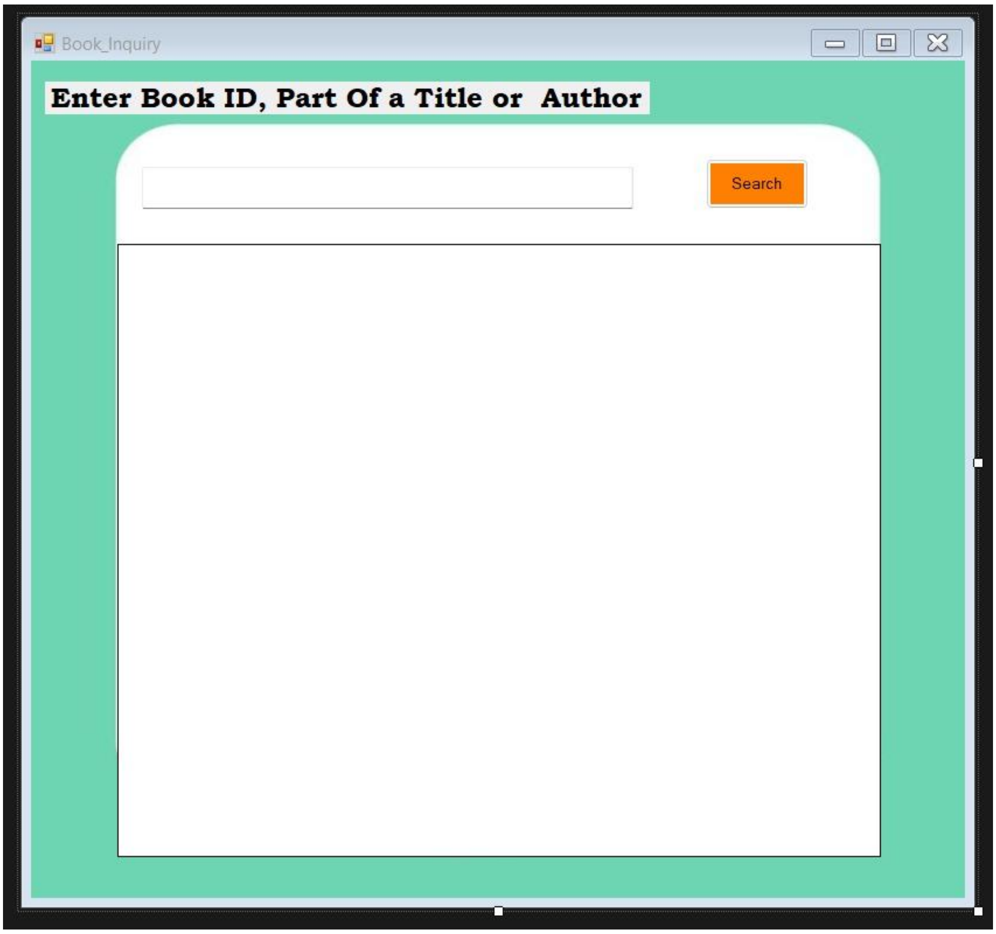
The Edit button brings an update in the master record of the chosen book title according to the given Book ID. This is an operation that makes changes to the available data in the database.

Delete Button

The Delete button deletes the selected book title in the database. Moreover, all the physical copy records and other related transaction data (loans and reservations, etc.) are also destroyed to preserve referential integrity.

Such format guarantees proper management of the collection of the library and consistency among related database records.

- Book Inquiry Form



The Book Inquiry Form is an interface with a specific search and exploration over the library collection. It enables the administrator to find book records fast depending on a variety of criteria.

User Interface Facilities and Elements.

Search Textbox

A search query in which the administrator will type a search query. The system allows searching on:

Book ID

Partial or full book title

Partial or full author name

Search Button

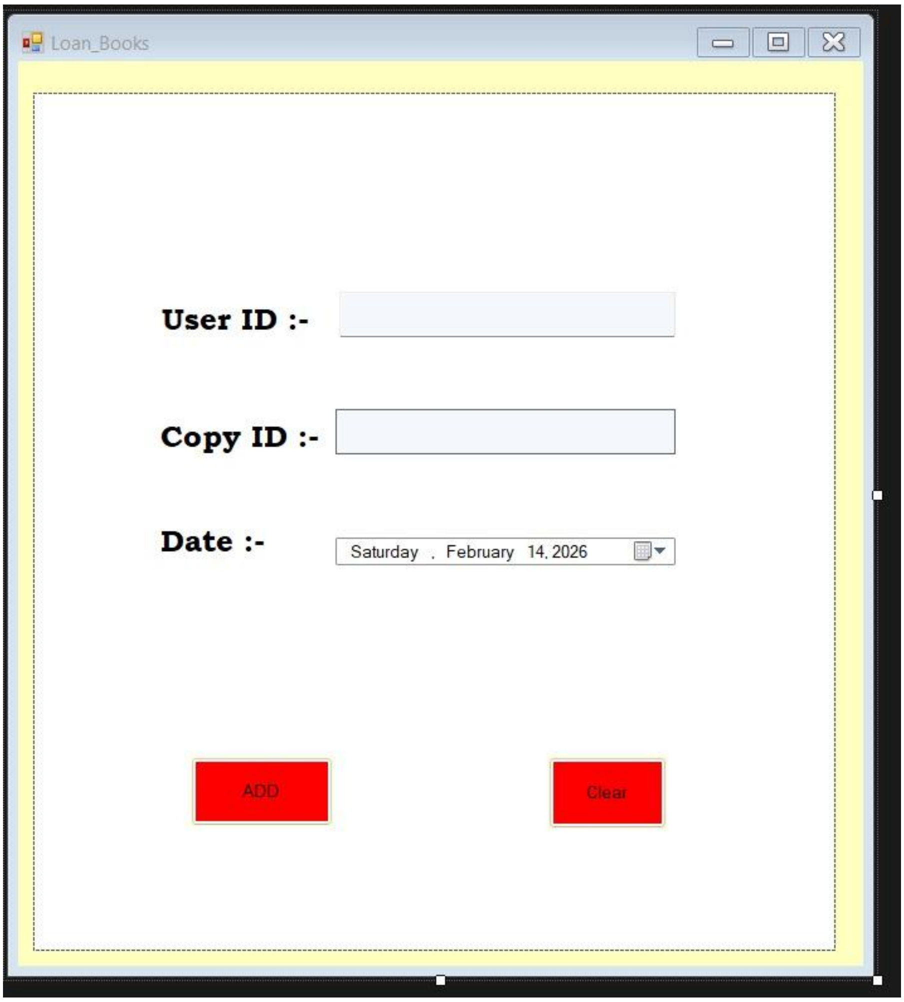
This button performs a database SELECT query and gets all the book records which satisfies the search criteria entered when this button is clicked.

Data Grid Panel

The search results are presented in a form of a structured data grid on a special panel. This enables the administrator to see a variety of records that match each other easily to take any additional action.

The Book Inquiry Form makes the system more user-friendly because it offers an effective and structured search system of finding particular books in the library inventory.

- Loan Books Form



The Book Loan Form This is a transactional interface that is used to handle the process of lending books to members of the library. It makes certain that there is accuracy in all loan operations and in compliance with library policies.

User Interface Components and Functionalities.

User ID and Copy ID Fields

Input boxes in which the administrator can specify:

Member who borrowed the book (User ID)

Copy ID is the physical copy of the book available in the loan.

Add Button

When enabled, this button carries out a number of validation tests, such as:

Ensuring that the member has already satisfied qualification to borrow (e.g. possesses less than five loans at any given time)

Making sure that the chosen copy is not reference only.

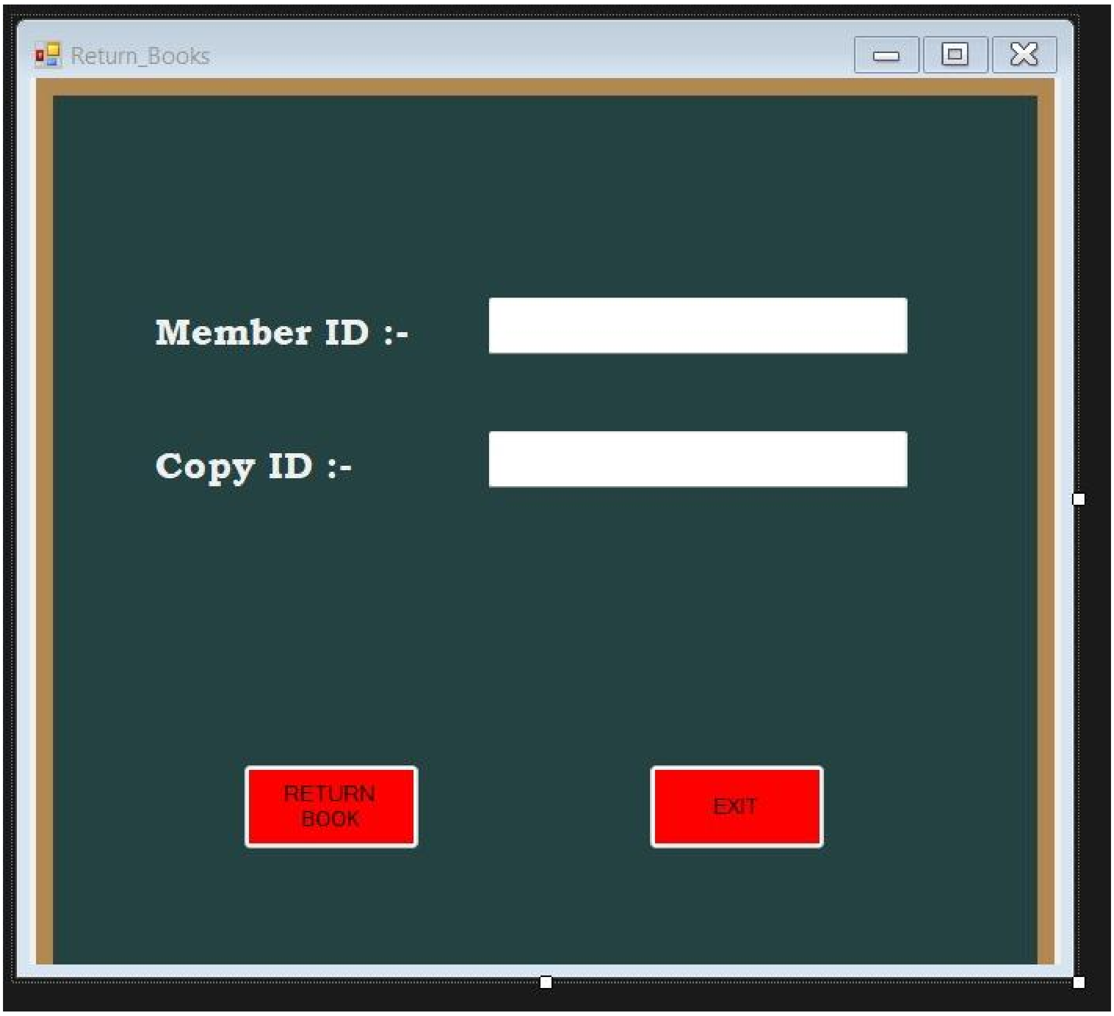
In case of success in the validation, the system:

Adds a new record in the Loans table.

Manages the status of the borrowed copy to show that it is already being loaned out.

This type of form centralizes and automatizes the process of loaning out and hence the library system is able to retain good record of the circulation of the books.

- Return Books Form



The Book Return Form is a dedicated transactional interface for managing the return of borrowed books. It ensures accurate updating of loan records and the availability status of returned copies.

User Interface Components and Functionalities

Copy ID Field

An input field where the administrator enters the **ID of the specific book copy** being returned. This identifies the exact copy in the system for processing.

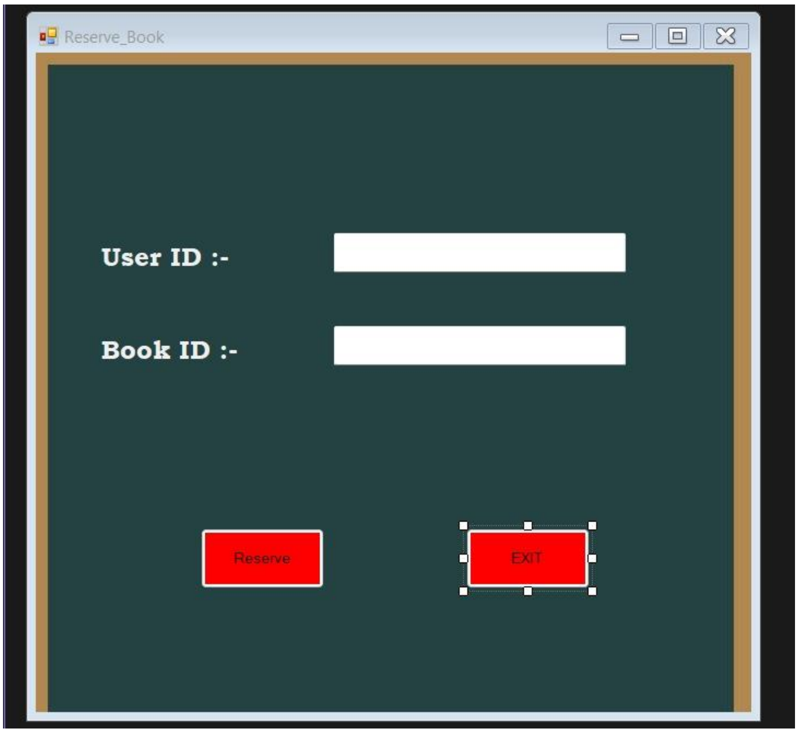
Return Button

When clicked, the system executes several key operations: Updates the corresponding **Loans** record to mark the transaction as **completed**.

Updates the **Copies** record to indicate the copy is now **available**. If a reservation exists for the same book, the status is updated to **reserved**, and the reservation queue is appropriately managed.

This form ensures efficient handling of book returns, maintains accurate inventory status, and respects the library's reservation policies.

- Reserve Books Form



The Book Reservation Form is a transactional interface that allows the administrator to record reservation requests from library members. It ensures that reservations are processed correctly and in accordance with the library's rules.

User Interface Components and Functionalities

User ID and Book ID Fields
Input fields that allow the administrator to specify:

- The **member** making the reservation (User ID)
- The **book title** being requested (Book ID)

Reserve Button

When activated, the system performs eligibility checks to confirm that the member is allowed to reserve the book. Upon successful validation, a new record is created in the **Reservations** table to track the member's request.

This form streamlines the reservation process, ensuring that members can request books efficiently while maintaining accurate tracking within the library system.

- View Users Form

The **View Users Form** is a dedicated, read-only interface designed to display a complete list of all patrons registered in the library system. It is accessed via the **"View User List"** button on the **User Operations** form. This form allows the administrator to quickly review user information without the distraction of editing controls.

User Interface Components and Functionalities

Data Grid View

The central and only functional component of the form is a data grid that presents all user records in a structured, tabular format.

- **Functionality:** When the form is loaded, the system automatically executes a SELECT query (e.g., SELECT * FROM Users) to retrieve all registered users from the database.
- **Behaviour:**
  - Each row in the grid corresponds to a single user.
  - Columns display key user information such as User ID, Name, NIC, Gender, Address, and User Type.

- The grid supports vertical scrolling, allowing the administrator to easily view all entries even when the user list is long.

This form provides a convenient, read-only overview of all library patrons, enhancing administrative efficiency and data accessibility.

8. Submit Software Development Project

The complete software project, including all C# source code, design files, and the database schema, is finalized and ready for submission.

User name– admin

Password- 12345

Drive Link-

https://drive.google.com/drive/folders/1QGdxgi9AtV1PkPcQvfUhh24K9kipVyJQ?usp=sharing